Graph-Based Multimodal Prognostic Modelling with Missing-Modality Robustness

Individual Project Pitch - Base Project: Kamineni, Multimodal Breast Cancer Diagnostic & Outcome Assistant

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Problem Statement

Kamineni's brief calls for diagnosis, risk stratification, and outcome prediction - this pitch takes the **outcome** half.

Two problems, addressed together:

1. **Prognosis $\neq$ classification.** Survival data is *censored*; accuracy and AUC are the wrong instruments. This requires time-to-event modelling and a **C-index**.
2. **Complete modalities are a laboratory fiction.** Genomic sequencing or high-resolution imaging is frequently missing in real practice; models trained assuming full availability degrade severely when a modality is absent.

Research Question.

Can a graph-structured fusion of tissue architecture and molecular pathways stratify survival risk - and does it degrade gracefully when modalities are missing or the cohort changes?

Data Availability

Hospital Data (Kamineni). Status: **Awaiting.** Brief notes existing internal AI work on breast cancer and outcomes.

Public Fallback - Required by Course Guidelines.

Development: TCGA-BRCA - WSI + RNA-seq + CNV + methylation + miRNA + clinical; survival labels (OS, PFI, censoring indicators) ship as standard, **no annotation needed**; open-access GDC tier covers everything required.

External: CPTAC-BRCA ($\sim$134 patients, adds proteomics) - free via the Proteomic Data Commons; enables a genuine cross-cohort survival test.

Graph Structure - Derived, Not Annotated.

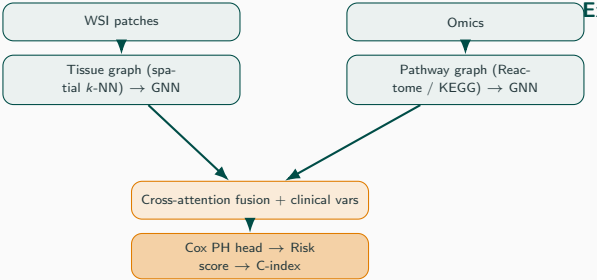
Pathway topology: Reactome v92 · KEGG · MSigDB Hallmark (50 gene sets, 4,383 genes)
Tissue graph: spatial k-NN over WSI patch coordinates

State of the Art / Relevant Work

Work	Relevance
MM-SurvNet (2024)	MaxViT + dual cross-attention; mean **C-index 0.64** on ER+ TCGA-BRCA (vs. image-only 0.53, clinical-only 0.47)
M3FusionNet (2026)	Reports the central failure mode: **13–22% cross-dataset AUC drop**
HEALNet (NeurIPS 2024)	Cross-attention fusion explicitly designed for **incomplete** biomedical modalities
M2EF-NNs (Sci. Rep. 2025)	Dempster–Shafer evidence theory for per-modality **uncertainty**
SELECTOR / HGM-SurvNet / cancer-type-aware fwk. (2024–26)	Modality-missing GNN / hypergraph baselines - **dropout-only** novelty
PathMoG (2026)	Pathway-centric GNN, 354 KEGG modules, 5,650 patients, 10 TCGA types

The Gap. Published survival models overwhelmingly assume complete modalities and single-cohort evaluation. Cross-cohort drop and missing-modality behaviour remain **open, not solved**.

Proposed Method & Expected Outcome



Modality dropout during training: randomly mask an entire modality (prob. p) to force graceful degradation.

Experimental Plan.

1. Baselines: clinical-only Cox, omics-only, WSI-only, naive concatenation
2. **Proposed**: graph fusion + modality dropout
3. Stress test at 20 / 50 / 80% modality-missing rates, with vs. without dropout training
4. TCGA → CPTAC vs. the reported 13–22% drop
5. Kaplan–Meier high-/low-risk curves + log-rank test

Metric. C-index (primary) - *not* accuracy (censored data).

Novelty. Tissue graph *combined with* a pathway-structured omics graph - differentiates from dropout-only baselines (SELECTOR / HGMSurvNet) and omics-only graphs (PathMoG).

Expected Outcome & Fallback. C-index vs. MM-SurvNet's 0.64 baseline · degradation curve vs. modality availability (central deliverable, rarely reported) · CPTAC cross-cohort C-index · Kaplan–Meier separation. **Fallback threshold:** if the tissue-graph GNN does not beat an attention-MIL survival baseline, drop it and lead with the pathway-omics-graph contribution.